27. Following the hint, we show that $H_{i-1}N$ is normal in H_iN. Let $h_{i-1}n_1 \in H_{i-1}N$ and $h_in_2 \in H_iN$ where the elements belong to the obvious sets. Using the fact that H_{i-1} is normal in H_i and that N is normal in G, we obtain $(h_in_2)h_{i-1}n_1(h_in_2)^{-1} = h_in_2h_{i-1}n_1n_2^{-1}h_i^{-1} = h_ih_{i-1}n_3n_1n_2^{-1}h_i^{-1} = h'_{i-1}h_in_4h_i^{-1} = h'_{i-1}h_ih_i^{-1}n_5 = h'_{i-1}n_5 \in H_{i-1}N$. Thus $H_{i-1}N$ is a normal sugroup of H_iN.

The hint does the rest of the work for us, except to observe at the end that H_i/H_{i-1} being simple implies that $[H_i \cap (H_{i-1}N)]/H_{i-1}$ is either trivial or isomorphic to H_i/H_{i-1}. Thus $(H_iN)/(H_{i-1}N)$ is either isomorphic to H_i/H_{i-1} or is trivial. Because $N = H_0N$ is itself simple, it follows at once that the distinct groups among the H_iN for $i = 0, 1, 2, \cdots, n$ form a composition series for G.

28. First we show that ψ is well defined. Let h_in_1 and h'_in_2 be the same elements of H_iN. Then $h_in_1 = h'_in_2$ so $h_i = h'_in_2n_1^{-1} = h'_in_3$. Because γ has kernel N, we see that $\psi(h_in_1) = \gamma(h_in_1)\gamma[H_{i-1}] = (h_iN)\gamma[H_{i-1}] = (h'_in_3N)\gamma[H_{i-1}] = (h'_1N)\gamma[H_{i-1}] = \gamma(h'_1n_2)\gamma[H_{i-1}]$, so ψ is well defined.

We show that ψ is a homomorphism. This follows from $\gamma(h_in_1h'_in_2) = \gamma(h_in_1)\gamma(h'_in_2)$ because γ is a homomorphism.

The kernel of ψ consists of all $x \in H_iN$ such that $\gamma(x) \in \gamma[H_{i-1}] = H_{i-1}N$ and ψ is clearly an onto map. By Theorem 34.2, $\gamma[H_i]/\gamma[H_{i-1}]$ is isomorphic to $(H_iN)/(H_{i-1}N)$. Exercise 27 shows that these factor groups are simple, and the desired result follows immediately.

29. Let $H_0 = e < H_1 < H_2 < \cdots < H_n = G$ be a composition series for G, and let $\phi : G \to G'$ be a group homomorphism of G onto G' with kernel N. Then $G' \simeq G/N$. Exercise 28 shows that the distinct groups among the groups H_iN for $i = 0, 1, \cdots, n$ form a composition series for G/N, and Exercise 27 shows that a factor group of this composition series is isomorphic to one of the factor groups in the composition series of groups H_i for G. Because G is a solvable group, it follows at once that all the factor groups in this composition series for G/N, composed of some of the groups H_iN, are also abelian, so that G/N is solvable

36. Sylow Theorems

1. 3 **2.** 27

3. A Sylow 2-subgroup of a group of order $24 = 8 \cdot 3$ has order 8. By Theorem 36.11, the number of them must be conguent to 1 modulo 2, and hence is an odd number. It must also divide 24, and the only odd divisors of 24 are 1 and 3 so the group has either one or three Sylow 2-subgroups.

4. The only numbers congruent to 1 modulo 3 that divide $255 = 3 \cdot 5 \cdot 17$ are 1 and $5 \cdot 17 = 85$. The only numbers congruent to 1 modulo 5 that divide 255 are 1 and $3 \cdot 17 = 51$.

5. Because $|S_4| = 24$, the Sylow 3-subgroups have order 3 and are thus cyclic and generated by a single 3-cycle. The possibilities are
$\langle(1, 2, 3)\rangle$,
$\langle(1, 2, 4)\rangle = \langle(3, 4)(1, 2, 3)(3, 4)\rangle$,
$\langle(1, 3, 4)\rangle = \langle(2, 4)(1, 2, 3)(2, 4)\rangle$, and
$\langle(2, 3, 4)\rangle = \langle(1, 4)(1, 2, 3)(1, 4)\rangle$.

6. A Sylow 2-subgroup of S_4 has order 8 and by Theorem 36.11, there must be either 1 or 3 of them. The group of symmetries of the square has order 8 and can be viewed as a subgroup of S_4 if we number the vertices 1, 2, 3, and 4.

If we number the four vertices in order 1, 2, 3, 4 counterclockwise, we obtain the group $H = \{(1), (1, 2, 3, 4), (1, 3)(2, 4), (1, 4, 3, 2), (1, 3), (2, 4), (1, 2)(3, 4), (1, 4)(2, 3)\}$.

If we number the four vertices in order 1, 3, 2, 4 counterclockwise, we obtain the group $K = \{(1), (1, 3, 2, 4), (1, 2)(3, 4), (1, 4, 2, 3), (1, 2), (3, 4), (1, 4)(2, 3), (1, 3)(2, 4)\}$.

If we number the four vertices in order 1, 3, 4, 2 counterclockwise, we obtain the group $L = \{(1), (1, 3, 4, 2), (1, 4)(2, 3), (1, 2, 4, 3), (1, 4), (2, 3), (1,3)(2, 4), (1, 2)(3, 4)\}$.

Because there can be at most three of them, we have found them all. We see that $K = (2, 3)H(2, 3)$ and $L = (3,4)H(3,4)$

7. The definition is incorrect. The order may be a power of p.

Let p be a prime. A ***p*-group** is a group with the property that the order of each element is some power of p.

8. The definition is incorrect. Elements of the normalizer are elements of G, not maps of G onto G.

The **normalizer** $N[H]$ of a subgroup H of a group G is the set of all $g \in G$ such that $\{ghg^{-1} \mid h \in H\} = H$.

9. The definition is misleading. A Sylow p-subgroup need not be unique, so replace the first two occurrences of "the" by "a". Also refer to the group as G after it is defined. The word "largest" also implies uniqueness, so replace it by "maximal".

Let G be a group whose order is divisible by a prime p. A **Sylow *p*-subgroup** of G is a maximal subgroup P of G with the property that P has some power of p as its order.

10. T T T F T F T T F F

11. *Closure:* Let $a, b \in G_H$. Then $aHa^{-1} = H$ and $bHb^{-1} = H$. Thus $(ab)H(ab)^{-1} = a(bHb^{-1})a^{-1} = aHa^{-1} = H$, so $ab \in G_H$.

Identity: For all $h \in H, ehe^{-1} = ehe = h$ so $eHe^{-1} = H$ and $e \in G_H$.

Inverses: Let $a \in G_H$. Then $aHa^{-1} = H$. Therefore $H = eHe = (a^{-1}a)H(a^{-1}a) = a^{-1}(aHa^{-1})a = a^{-1}Ha$, so $a^{-1} \in G_H$. Thus G_H is a subgroup of G.

12. Let H be a Sylow p-subgroup of G. Because q divides $|G|$, we know that $H \neq G$. For each $g \in G$, the conjugate group gHg^{-1} is also a Sylow p-subgroup of G. Because G has only one Sylow p-subgroup, it must be that $gHg^{-1} = H$ for all $g \in G$, so that H is a proper normal subgroup of G, and G is thus not a simple group.

13. The divisors of 45 are 1, 3, 5, 9, 15, and 45. Of these, only 1 is congruent to 1 modulo 3, so by Theorem 36.11, there is only one Sylow 3-subgroup of a group of order 45. By the argument in Exercise 12, this subgroup must be a normal subgroup.

14. Let G be a p-group, so that by definition, every element of G has order a power of p. If a prime $q \neq p$ divides $|G|$ then G has an element of order q by Cauchy's theorem, contradicting that G is a p-group. Thus the order of G must be a power of p.

Conversely, if the order of G is a power of p, then the order of each element of G is also a power of p by the Theorem of Lagrange. Thus G is a p-group.